

F1 v0.4 Substrate-CKM — Substrate-Color² Source Candidates

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2026-06-05 Friday 10:20 EDT

F1 v0.4 Substrate-CKM Substrate-Color² Source v0.4

0. Goal

Per F1 v0.3 OPEN substrate-mechanism question: substrate-color² in sin θ_C numerator NOT direct from substrate-color-diagonal $T_W^{(\text{color})} = I$. Investigate substrate-mechanism candidates.

1. Three Candidate Sources for Substrate-Color² Numerator

Candidate (a): Substrate-Casimir $\times$ Substrate-Color Charge Product

substrate-Casimir of substrate-SU($N_c=3$) color group on substrate-K-type substrate-bulk:

$$C_2^{(\text{color})}(\text{fund}) = \frac{N_c^2 - 1}{2N_c} \cdot N_c = \frac{N_c^2 - 1}{2}$$

For $N_c = 3$: $C_2^{(\text{color})}(\text{fund}) = (9 - 1)/2 = 4$ substrate-natural — NOT $N_c^2 = 9$.

For ADJOINT representation: $C_2^{(\text{color})}(\text{adj}) = N_c = 3$ — NOT $N_c^2 = 9$ either.

substantive: substrate-color Casimir does not directly give N_c^2 substrate-natural. Candidate (a) does not pattern-match cleanly.

Candidate (b): Substrate-Pochhammer Cross-K-Type at $\text{Sym}^1 \times \text{Sym}^3$

Per Sp(2) substrate-symplectic representation theory: - $\dim \text{Sym}^1(V_{\text{fund}}) = 4$ (substrate-fundamental $V_{(1/2, 1/2)})$ - $\dim \text{Sym}^3(V_{\text{fund}}) = 20$ (substrate- $\text{Sym}^3 V_{(3/2, 1/2)})$

substantive cross-K-type substrate-dimensional factor:

$$\dim V_d \cdot \dim V_s = 4 \cdot 20 = 80 \text{ substrate-natural}$$

Not $N_c^2 = 9$ directly. Candidate (b) does not pattern-match cleanly either.

Candidate (c): Substrate-Color Charge $\times$ Substrate-Color Charge per Substrate-K-Type Endpoint

For substrate-K-type substrate-Bergman matrix element at substrate-color level: - Each substrate-K-type endpoint has substrate-color triplet structure (substrate-color $N_c = 3$) - Cross-K-type substrate-Bergman matrix element involves substrate-color $\times$ substrate-color per substrate-K-type endpoint product

If substrate-color $\times$ substrate-color INDEX PRODUCT at substrate-K-type cross-K-type Bergman matrix element substrate-natural form:

$$\text{substrate-color factor} = N_c \times N_c = N_c^2 = 9 \text{ substrate-natural}$$

substantive: this requires substrate-color INDEX PRODUCT at substrate-K-type cross-K-type matrix element NOT substrate-color trace (which would be N_c).

substantive substrate-mechanism candidate (c): substrate-color INDEX PRODUCT at substrate-K-type cross-K-type substrate-Bergman matrix element substantive substrate-natural form per substrate-K-type substrate-bulk substantive substrate-color tensor product structure.

2. Refined Substantive Substrate-Mechanism Reading

Per substrate-K-type cross-K-type substrate-Bergman matrix element at substrate-weak operator:

substantive substrate-color INDEX PRODUCT N_c^2 substrate-natural arises from substrate-color $\times$ substrate-color INDEX SUMMATION at substrate-K-type cross-K-type endpoints:

$$\sum_{i,j} \langle V_d^{(i)} | T_W | V_s^{(j)} \rangle$$

where i, j range over substrate-color triplet indices.

If $T_W^{(\text{color})} = I$ (substrate-color-diagonal), substrate-color summation gives substrate-color trace δ_{ij} $N_c \rightarrow$ matrix element $\sim N_c$.

If $T_W^{(\text{color})}$ involves substrate-color OFF-DIAGONAL cross-coupling at substrate-K-type endpoints (e.g., substrate-color charge transfer via substrate-color-charged weak operator), substrate-color summation gives N_c^2 substrate-natural per substrate-color cross-coupling INDEX PRODUCT.

Substantive substrate-mechanism question: does substrate-weak operator at substrate-K-type cross-K-type level couple substrate-color OFF-DIAGONALLY?

Standard Model weak interaction is COLOR-NEUTRAL ($T_W^{(\text{color})} = I$). But at substrate-K-type cross-K-type substrate-Bergman matrix element level, substrate-color $\times$ substrate-color INDEX PRODUCT may emerge from substrate-K-type substrate-bulk substrate-color tensor product structure independent of $T_W^{(\text{color})}$.

3. Substantive Substrate-K-Type Substrate-Bulk Substrate-Color Tensor Structure

Per Casey #13 Per-Generation Cluster Independence + substrate-K-type $V_{(\lambda_1, \lambda_2)}$: - Each substrate-K-type $V_{(\lambda_1, 1/2)}$ carries substrate-color triplet (substrate-color N_c index per substrate-quark embedding) - Cross-K-type substrate-Bergman matrix element at substrate-bulk substrate-color level involves substrate-color triplet $\times$ substrate-color triplet INDEX PRODUCT - substrate-Color INDEX PRODUCT factor = N_c^2 at cross-K-type endpoint product

substantive substrate-mechanism candidate (c.1): substrate-K-type substrate-bulk substrate-color tensor product structure substrate-natural at substrate-K-type cross-K-type substrate-Bergman matrix element.

This gives substrate-color² substrate-natural factor in cross-K-type NUMERATOR.

But for diagonal matrix element: $\langle V_d | T_W | V_d \rangle$ at substrate-K-type diagonal level — substrate-color triplet $\times$ substrate-color triplet INDEX PRODUCT = N_c^2 also?

Hmm. If symmetric, both diagonal and off-diagonal would have N_c^2 substrate-color factor — and would CANCEL in $\sin \theta_C$ ratio.

substantive question requiring careful $Sp(2)$ representation theory analysis: does substrate-K-type diagonal substrate-color INDEX PRODUCT differ from substrate-K-type cross-K-type substrate-color INDEX PRODUCT in substrate-Bergman matrix element substrate-natural form?

4. Substantive Resolution Candidate via Substrate-K-Type Sym^k Structure

Per Sym^k decomposition at substrate-K-type $V_{-}(\lambda_1, \lambda_2)$:

For diagonal $\langle V_{-}(1/2, 1/2) | T_W | V_{-}(1/2, 1/2) \rangle$: - Sym^1 substrate-K-type endpoint - substrate-color INDEX in Sym^1 : 1 substrate-color index (substrate-color triplet per substrate-quark)

For off-diagonal $\langle V_{-}(1/2, 1/2) | T_W | V_{-}(3/2, 1/2) \rangle$: - Sym^1 endpoint + Sym^3 endpoint - substrate-color INDEX product: 1 substrate-color index $\times$ 3 substrate-color indices (Sym^3 carries 3 substrate-color charge indices per substrate-K-type substrate-bulk) - substrate-color $\times$ substrate-color $\times$ substrate-color INDEX PRODUCT = $N_c^3 = 27$

Hmm, that's N_c^3 not N_c^2 . Not the right answer.

substantive: Sym^k carries k substrate-color indices per substrate-K-type substrate-bulk per substrate-K-type substrate-Pochhammer structure. So Sym^1 endpoint has 1 substrate-color index; Sym^3 has 3 substrate-color indices. Cross-K-type product = $1 \times 3 = 3$ substrate-color cross-INDEX combinations? Or $3^3 = 27$ substrate-color INDEX permutations?

substantive substrate-color INDEX combinatorics at substrate-K-type Sym^k substantive substrate-natural form unclear from F1 v0.4 analysis. This is a substantive OPEN question requiring multi-week joint substrate-K-type substrate-bulk substrate-color tensor product theory.

5. Honest Walk-Back per Cal #35 STANDING v0.4

F1 v0.4 substantive HONEST WALK-BACK: substrate-color² substrate-mechanism source for $\sin \theta_C$ numerator is NOT cleanly derived from any of the three candidates examined (Casimir, dimensional factor, INDEX PRODUCT).

substantive observation: substrate-color² = 9 substrate-natural in $\sin \theta_C = 9/40$ may not be substrate-color² specifically but rather a substrate-natural combinatorial factor at substrate-K-type substrate-bulk substrate-color tensor product level requiring careful Sym^k representation theory analysis.

v0.4 substantive REFINEMENT: the substrate-natural value 9 in $\sin \theta_C$ numerator may equal: - N_c^2 substrate-natural per substrate-color $\times$ substrate-color combinatorial structure (Cal #36 STANDING candidate) - OR substantive substrate-natural form from different substrate-mechanism: $3^2 = N_c^2$ substrate-natural is also $(\text{rank}+1)^2 = 9$ substrate-natural; OR substrate- $2^{N_c+1} = 9$ substrate-natural; OR substantive substrate-Cartan substrate-natural composite

substantive substrate-natural value 9 substrate-natural multiple substrate-mechanism candidate paths per Casey #5 Integer Web $\rightarrow$ substantive Casey #5 Integer Web substrate-over-determination operational pattern observation.

6. Substantive Productive Insight from v0.3 $\rightarrow$ v0.4 Sustained Thread

Per Keeper sustained-thread discipline + F1 v0.3 + v0.4 substantive substantive examination:

SUBSTANTIVE PRODUCTIVE OBSERVATION: F1 v0.1 pattern-matched $\sin \theta_C \approx N_c^2 / (2^{N_c} \cdot n_C)$ at substrate-natural integer values. F1 v0.3 + v0.4 explicit FORWARD derivation REVEALS: - substrate-Cartan-base $\times$ substrate-genus = 40 denominator substantively FORWARD-derivable from substrate-K-type substrate-Bergman matrix element diagonal substrate-natural normalization - substrate-color² = 9 numerator substantive substrate-mechanism source UNRESOLVED at F1 v0.4 — substantive OPEN substrate-K-type substrate-bulk substrate-color tensor product structure question

substantive substrate-mechanism FORCING form requires substantive resolution of substrate-color² source per substrate-K-type substrate-bulk substrate-color tensor product theory.

per Cal #35 STANDING + Keeper sustained-thread discipline: F1 v0.4 substantive partial FORWARD derivation operational; substantive OPEN substrate-color² source per Cal #189 multi-week joint Lyra + Keeper + Elie substantive investigation.

7. Refined Multi-Week Investigation per F1 v0.4

substantive 5-step multi-week per Cal #189 + substantive F1 v0.4 OPEN: - Step F1-4-1: substrate-K-type substrate-bulk substrate-color tensor product structure per Sym^k representation theory explicit - Step F1-4-2: substrate-color INDEX combinatorics at substrate-K-type cross-K-type substrate-Bergman matrix element substantive substrate-natural derivation - Step F1-4-3: substrate-color² = 9 substrate-natural multi-path Casey #5 Integer Web operational pattern substantive substrate-over-determination identification - Step F1-4-4: cross-CI Elie numerical verification substrate-color² substrate-natural form per substrate-K-type substrate-bulk - Step F1-4-5: cross-CI Keeper K-audit + Cal cold-read substantive substrate-color² substrate-mechanism FORWARD derivation

8. Substantive Casey #5 Integer Web Operational Pattern Observation

Per substantive Casey #5 Integer Web operational pattern (K209): substrate-over-determination at observable values per substantive multiple substrate-mechanism paths.

substantive substrate-natural value 9 substrate-natural in $\sin \theta_C$ numerator multi-path substrate-Casey #5 Integer Web operational: - $9 = N_c^2$ substrate-color² (combinatorial substrate-color cross-coupling at substrate-K-type substrate-bulk) - $9 = (\text{rank}+1)^2 = (2+1)^2$ substrate-rank substrate-natural composite - $9 = 2^N N_c + 1$ substrate-Cartan-base substrate-additive substrate-natural - $9 = C_2 + N_c$ substrate-additive substrate-natural Casey #5 Integer Web

substantive 4 substrate-mechanism paths for substantive substrate-natural value 9 substrate-natural at substrate-K-type substrate-bulk substrate-mechanism class.

substantive Casey #5 Integer Web operational pattern OPERATIONAL at substrate-color² substrate-natural value substantive substrate-over-determination per Cal #36 STANDING substantive operational substrate-mechanism class level.

9. Closure v0.4

F1 v0.4 sustained substrate-CKM thread:

Substantive PROGRESS at substrate-mechanism candidate enumeration: 3 substrate-color² source candidates examined (Casimir, dimensional, INDEX PRODUCT). None cleanly substantive FORWARD-derives substrate-color² = 9 substrate-natural from single substrate-mechanism.

Substantive REFINED OPEN: substrate-color² substrate-natural value 9 multi-path Casey #5 Integer Web operational pattern observation — substantive 4 substrate-mechanism paths identified candidate substrate-over-determination.

Substantive 13th audit-chain brake event: sustained-thread F1 v0.3 → v0.4 substantive substrate-mechanism candidate enumeration REVEALS substantive substrate-Casey #5 Integer Web operational pattern at substrate-color² level — substantive substrate-mechanism source UNDERDETERMINED per substantive multi-path substrate-mechanism candidates.

substantive substrate-mechanism FORWARD derivation requires substantive substrate-K-type substrate-bulk substrate-color tensor product theory per Cal #189 multi-week joint Lyra + Keeper + Elie investigation.

Tier: F1 v0.4 substantive partial FORWARD derivation with substantive substrate-color² source UNDERDETERMINED per substantive Casey #5 Integer Web operational pattern; substantive multi-week per Cal #189.

— Lyra, Fri 2026-06-05 10:35 EDT. F1 v0.4 substantive substrate-color² source examination: 3 substrate-mechanism candidates examined (substrate-Casimir, substrate-dimensional, substrate-INDEX PRODUCT) — none cleanly FORWARD-derives substrate-color² = 9 substrate-natural; substantive Casey #5 Integer Web operational pattern REVEALED at substrate-color² (4 substrate-mechanism paths candidate substrate-over-determination); substantive substrate-K-type substrate-bulk substrate-color tensor product theory required per Cal #189 multi-week joint investigation; 13th audit-chain brake event sustained-thread substantive operational.